

Purpose

To select allow MRP users to send certain reports e.g. invoices as e-mails to the customer.

Needs to have some sort of “sent” functionality and handle errors sending e-mails.

The E-mail System

There are some technical limitation to achieving this end notably:

- MRP clients are on a closed IP domain with no outside network access
- MRP clients have no access to their own hard drive or USB devices
- Need to decide, do we send the e-mails from the user PC or a central source?

First change made was to implement virtual machines for the MRP clients in Sales.

This now allows a folder to be created on the “host” PC in this case:

My documents\PDFs_from_MRP

This folder is shared with the virtual machine as a network path.

Phew!

Now to get the PDF created BullZip was installed on the clients virtual MRP machine, then the invoicing screen was modified to allow PDF invoicing.

Here is a picture of the existing invoicing screen:

	Sales Order ID	Cust PO ID	Customer ID	Ship Date
Process	244544	42473	S443	11/11/2021 15:00:
From:	244578	01027	S643	11/11/2021 14:59:
	244598	bir646050	s2004	11/11/2021 15:02:
Process	244611	BIR646055	S2004	11/11/2021 15:02:
From:	244673	PPE/45886	S1334	11/11/2021 14:43:
	244741	272-075444 X	S1540	11/11/2021 14:34:

The screen uses a combobox to show the sales orders available to invoice, this control can sometimes be slow to display data depending on system load and the amount of records it needs to show.

E-mail from MRP Design Specification

Created By: Roger Williams

The new screen looks like this:

SO ID	Cust ID	Customer
☐ 244143	S902	Gil-Lec Limited
☐ 244590	s0001	Utility Account
☐ 244593	S2004	Contact Electrical Wholesale Limited
☐ 244601	s2004	Contact Electrical Wholesale Limited
☐ 244602	S1294	Templegate Electrical Supplies Limited
☐ 244603	s2004	Contact Electrical Wholesale Limited
☐ 244623	S888	City Electrical Factors (Nelson Group)
☐ 244643	388	Bernlite Limited
☐ 244681	S1190	Sunny Electrical Supplies

This uses a listview to show the sales orders that can be invoiced. Also it allows the user to select all or individual sales orders, plus invoices can be just printed, printed to PDF, e-mailed, or printed *then* e-mailed.

When this process is run the PDFs are created in the shared folder on the “host” PC, this ends the first part of the process.

The PDFs are created using a standard naming convention, also an accompanying text file is also created.

Filename format:

<CustomerID> & "_" & <Customer name> & "_SalesOrder_" & <Sales order number> & "_Invoice_" & <Invoice number> & "_" & <Date in format: "ddmmyyyy"> & "_" & <Time> & "." & <File extension e.g. PDF/TEXT>

A text file is created with the same name format this contains:

- PDF filename
- E-mail subject e.g. Invoicing
- E-mail address to send to
- MRP user name
- Sales order region type e.g. UK/Non EU

The shared folder has these sub folders:

- Sent
- FailedEmails

An external PC was configured to act as an e-mail “server”, on this PC a VB .Net program was written using Visual Studio 2015 to perform all the necessary copying/e-mailing.

A program is run every 40 minutes by the Windows Task Scheduler, this checks the shared folder on the PCs marked for e-mailing for anything to send. The program checks a number of predefined PCs for files to e-mail.

E-mail from MRP Design Specification

Created By: **Roger Williams**

The e-mail “server” has these folders:

- Sent
- Failed
- UserDeleted

When the program finds files to e-mail it:

- Edits the TXT file and add the users PC name to it
- Copies them into the Sent folder on the users PC
- Moves them to the emails_from_MRP folder on the e-mail “server”

Another program run every 60minutes by the Windows Task Scheduler, e-mails the files.

It reads the TXT to use as abase for e-mail as I contains who too, who from and what to send.

The sales region type is used to determine which HTML template is used as the message body, as UK and Non EU sales have different conditions of sale. Plus it allows for the company Logo/affiliations etc. to be shown on the e-mail.

If the e-mail sends, a log is written on the e-mail “server” and an e-mail is sent to the user who sent the files for e-mailing telling them which files were sent.

The files are then moved into the Sent folder under the users name in a folder with that days date formatted YYYY_MM_DD e.g.:

Sales

Bob Scratchit\2021_11_15

Jone Doe\2021_10_12

IF the e-mail didn’t send, an error log is written on the e-mail “server” into the “Failed” folder, containing the PDF filename the date/time and SMTP error message.

Then an e-mail is sent to user containing this log, and this log is also copied into the users “FailedEmails” folder.

The files to e-mail stay in the “live” e-mails folder until they can be successfully sent.

To aid the user in correcting possible e-mail address errors in failed e-mails, a text editor exists on each users PC, the design of which looks like this:

The screenshot shows a Windows-style application window titled "Email From MRP Text Editor". It has a standard Windows title bar with minimize, maximize, and close buttons. The window is split into two panes. The left pane, titled "Failed Email Folder Contents", contains a label "File Name" above a large, empty text area. Below this text area is a button labeled "Delete File". The right pane, titled "Text File Contents", contains several labels: "PDF Filename:", "Email Type:", "Email To:", "Email From:", "Order VAT Territory:", and "Email Request PC:". The "Email To:" label is followed by a text input field. Below these labels is a button labeled "Save Changes". At the bottom right of the window is a button labeled "Close".

This edits the TXT file on the e-mail “server” directly by *only* allows e-mail address modification.